

Notes: *Stentor coeruleus* Photokinesis and the Action-Spectrum Literature

General background notes on photokinesis in *Stentor coeruleus*, built from Iwatsuki (1992) plus everything else we've since pulled in the same literature. For the actionable “which wavelength should we test” recommendation, see `WAVELENGTH_SELECTION_GUIDE.md` — this document is the background/citation-heavy companion to that decision doc.

Background: three photoresponse types

S. coeruleus has three distinct light responses:

- **Photophobic response** — a sudden increase in light causes a stop/ciliary-reversal (“avoiding reaction”).
- **Phototaxis** — directed swimming away from a collimated light source (negative phototaxis).
- **Photokinesis** — steady-state swimming speed changes with ambient light level (faster in light, slower in dark; “positive photokinesis”).

Iwatsuki (1992) reports *S. coeruleus* as the first protozoan shown to exhibit all three. The photophobic response and phototaxis are well established to share a single photoreceptor: **stentorin**, a hypericin-based chromophore concentrated in pigment granules under the cell membrane, with absorption confined to roughly 525-630 nm. That absorption band is also *why* the cell looks blue-green: perceived pigment color is the complement of what's absorbed, so absorbing green-through-red leaves blue reflected back.

Iwatsuki additionally claims photokinesis has its *own*, separate action spectrum peaking at 680 nm — well outside stentorin's absorption band — which would imply a second, distinct photoreceptor system. That claim is the main thing this document interrogates, because our own StentorCam data shows little to no photokinetic response at 680 nm.

Iwatsuki (1992): what it reported

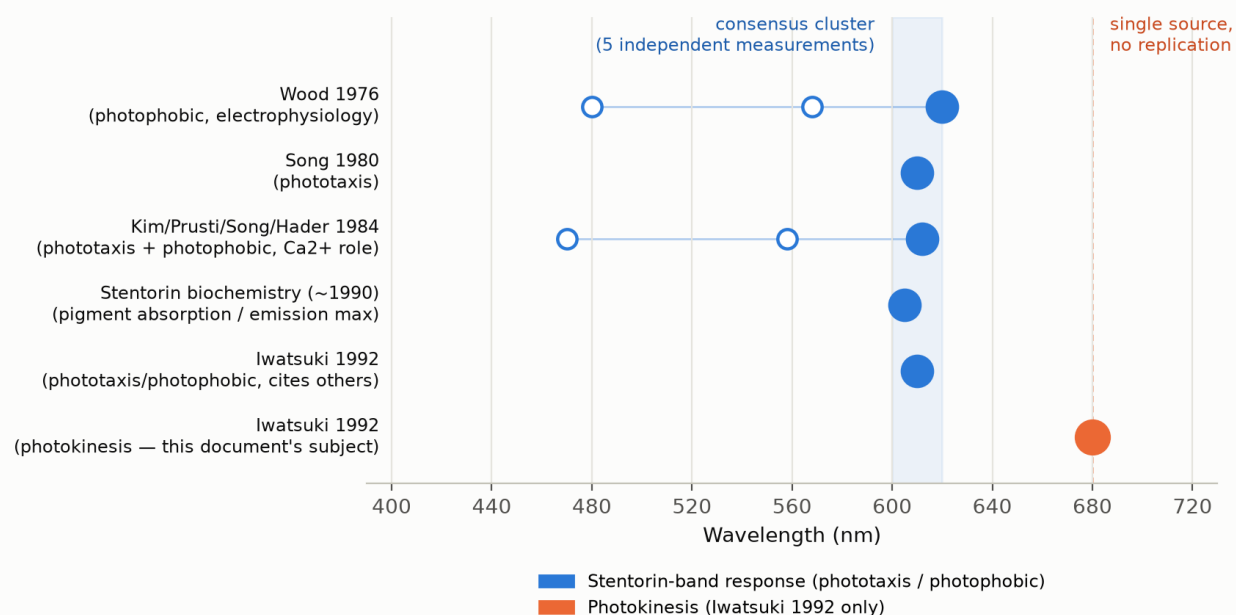
Source: Iwatsuki, K. (1992). *Stentor coeruleus* shows positive photokinesis. *Photochemistry and Photobiology*, 55(3), 469-471. (PDF: `references/Iwatsuki-1992-Stentor-coeruleus-positive-photokinesis.pdf`)

- Mean swimming velocity ~0.6 mm/s at 100 lx, up to ~1.0 mm/s at 50000 lx (white light).
- Velocity increases roughly **linearly with log(fluence rate)** — a Weber-Fechner-type relationship, not linear in raw light intensity.
- The claimed photokinesis action spectrum (their Figure 3) peaks at **680 nm**, built by measuring a fluence-rate-vs-velocity dose-response curve at each of 9 wavelengths (400-720 nm) and reading off the fluence rate needed to reach a fixed criterion velocity (0.58 mm/s) at each one — collapsing 9 dose-response curves into 9 comparable numbers, effectively an EC50-per-wavelength. (The mechanics of reading that figure — axis units, why 0.58 mm/s — are covered in the appendix at the bottom of this document, since they're specific to this one paper's figure rather than general findings.)
- Their light source was a filtered incandescent lamp (interference filters, <6 nm half-bandwidth, plus neutral-density filters) — see the light-source comparison table in `WAVELENGTH_SELECTION_GUIDE.md` for how this compares across the whole literature.

Cross-paper wavelength synthesis

Pooling every reported action-spectrum/absorption peak across the *S. coeruleus* literature we've gathered (not just Iwatsuki) makes the pattern clear:

Reported action-spectrum peaks across *Stentor coeruleus* photoresponse literature



Source	Response type	Reported peak(s)
Wood 1976	Photophobic (electrophysiology)	620 nm primary; 480 nm shoulder; 568 nm secondary (absorption data only)
Song 1980	Phototaxis	~610 nm (broad maximum)
Kim/Prusti/Song/Hader 1984	Phototaxis + photophobic, Ca2+ role; quantum-efficiency action spectrum vs. absorption	~610-615 nm primary; ~555-560 nm secondary; ~470 nm shoulder
Stentorin biochemistry (~1990, Song lab)	Pigment absorption / fluorescence-emission max	~600-620 nm
Iwatsuki 1992	Phototaxis/photophobic (cites the above)	610 nm
Iwatsuki 1992	Photokinesis	680 nm — the outlier

Values for Wood 1976, Song 1980, and Kim/Prusti/Song/Hader 1984 confirmed directly from the primary-source PDFs (kept in a local reference library, not committed here — see `WAVELENGTH_SELECTION_GUIDE.md`'s Source Status section for why); only the stentorin-biochemistry summary is still secondary, pending access.

Five independent measurements, spanning 1976-1992 and multiple labs/methods (behavior, electrophysiology, pigment biochemistry), cluster tightly at 600-620 nm. Averaging the five primary peaks (620, 610, 612, ~605, 610) gives **~612 nm** as the consensus wavelength — the single best bet for provoking *any* photoresponse in *S. coeruleus*, stentorin-mediated or not. Wood's secondary features (568/480 nm) are independently corroborated by Kim/Prusti/Song/Hader's own action spectrum (~555-560/470 nm) — two labs, two different methods, converging on the same multi-peak shape, which is much stronger evidence than a single paper. Still, treat those secondary peaks as lower-confidence than the 610-620 nm primary cluster: Wood's 568 nm point specifically was never directly tested in his action spectrum (only inferred from absorption spectra), unlike Kim/Prusti/Song/Hader's ~555-560 nm point, which was directly measured.

680 nm is the outlier by every measure here: it's the only point off the cluster, it comes from a single paper, and — as the next section covers — it has no matching pigment and has never been independently replicated.

Why the 680 nm photokinesis peak looks weak

Citation history: cited as background, never independently re-tested

Iwatsuki (1992) has been cited at least 24 times (per Semantic Scholar), almost entirely in reviews and later work on *Stentor/Blepharisma* photobiology — e.g. Fabczak and coworkers’ photosensory-transduction series (1993, 2006, 2008, 2010), the *Photomovements in Ciliated Protozoa* book chapters (1998, 2001), and a 2022 paper on behavioral-mode switching in *S. coeruleus*. In every citing paper we found, the 680 nm photokinesis action spectrum is repeated as established background alongside the much better-corroborated 610 nm phototaxis/photophobic-response peak — none appear to independently re-measure or challenge it.

No known pigment absorbs at 680 nm in this organism

Stentorin’s absorption is confined to roughly 525-630 nm (Song 1980; Wood 1976; the stentorin structure papers, Tao/Ghetti/Song ~1990) — essentially none at 680 nm. Iwatsuki’s own argument for “a second photoreceptor system” rests entirely on 680 nm not fitting stentorin’s absorption at all — but in the 30+ years since, no one appears to have identified or proposed a candidate pigment that actually absorbs at 680 nm in this non-photosynthetic ciliate (*S. coeruleus* has no plastids/chlorophyll). The claim introduces an unidentified photoreceptor with no follow-up characterization work.

An independent, broader check points the same way: an AI-assisted literature synthesis (Consensus.app, run July 2026, screening roughly 150 papers across the wider ciliate photoreception literature — kept in a local reference library, not committed here for copyright reasons) concluded that “the included literature does not provide strong direct evidence for a second established visible-light photoreceptor alongside stentorin” in *S. coeruleus* specifically. Its own comparative caveat — that other ciliates/heterotrichs can use non-hypericin photoreceptors (rhodopsins, flavins) — is explicitly flagged in that source as a cross-species point, not evidence for a second pigment within *S. coeruleus* itself.

Conclusion

The 610 nm peak (phototaxis/photophobic response) is corroborated by multiple independent groups using absorption spectroscopy, electrophysiology, and behavior. The 680 nm photokinesis peak, by contrast, rests on this single 1992 paper: no independent replication found, no identified photoreceptor pigment to explain it, and (see the appendix) a methodologically thin criterion-velocity derivation. That gives a reasonable literature-grounded basis for our data showing little response at 680 nm — we may be observing a real discrepancy with an isolated, potentially non-reproducible historical result, rather than an artifact of our own setup.

Kühnel-Kratz & Häder (1994) follow-up: no wavelength retest, but two other caveats

Full text obtained (kept in a local reference library, not committed — see `WAVELENGTH_SELECTION_GUIDE.md` Source Status). It does **not** retest the 680 nm question — their light-reaction experiments used white light only, no monochromatic wavelength sweep, so it neither confirms nor refutes Iwatsuki’s action spectrum. Two other findings from it are still relevant to our own setup, though:

- They observed positive photokinesis **only at fluence rates below $\sim 10 \text{ W/m}^2$** ; above that they saw no clear kinesis, in tension with Iwatsuki’s claim of an almost-linear velocity-vs-log-fluence relationship across the *entire* tested range. Worth checking where our own fluence rates fall relative to this 10 W/m^2 cutoff.
- Their dark-swimming baseline velocities (1.5-1.8 mm/s) were notably higher than every previously published value (0.2-1.2 mm/s, including Iwatsuki’s), which they attribute to their wider 25 mm cuvette reducing wall effects seen in smaller dishes/slides. If our chamber geometry is closer to the older, narrower setups, our baseline (dark) velocity may not be directly comparable to theirs either — worth controlling for chamber width/depth when comparing absolute velocities across studies.

Kim/Prusti/Song/Hader (1984): obtained via interlibrary loan

Full text obtained via interlibrary loan (kept in a local reference library, not committed — see WAVELENGTH_SELECTION_GUIDE.md Source Status). Its Fig. 1 plots a proper quantum-efficiency action spectrum for the step-up photophobic response directly against stentorin’s measured absorption spectrum (both in vivo and at 77 K) — a more direct optical comparison than Iwatsuki’s inverted-fluence-rate “relative response” proxy (see appendix). The action spectrum shows the same multi-peak shape as Wood 1976: primary peak ~610-615 nm, secondary ~555-560 nm, smaller shoulder ~470 nm, then a sharp drop-off past ~630-650 nm. Light source: a 300 W slide projector (Gold) with ~10 nm half-bandwidth interference filters and a Kettering radiant power meter — the same lamp-and-filter pattern as every other paper in this literature (see the light-sources comparison in WAVELENGTH_SELECTION_GUIDE.md).

Appendix: how Iwatsuki’s Figure 3 was actually built

Detail specific to this one paper’s figure, kept here rather than in the main body above since it’s methodological mechanics rather than a general finding.

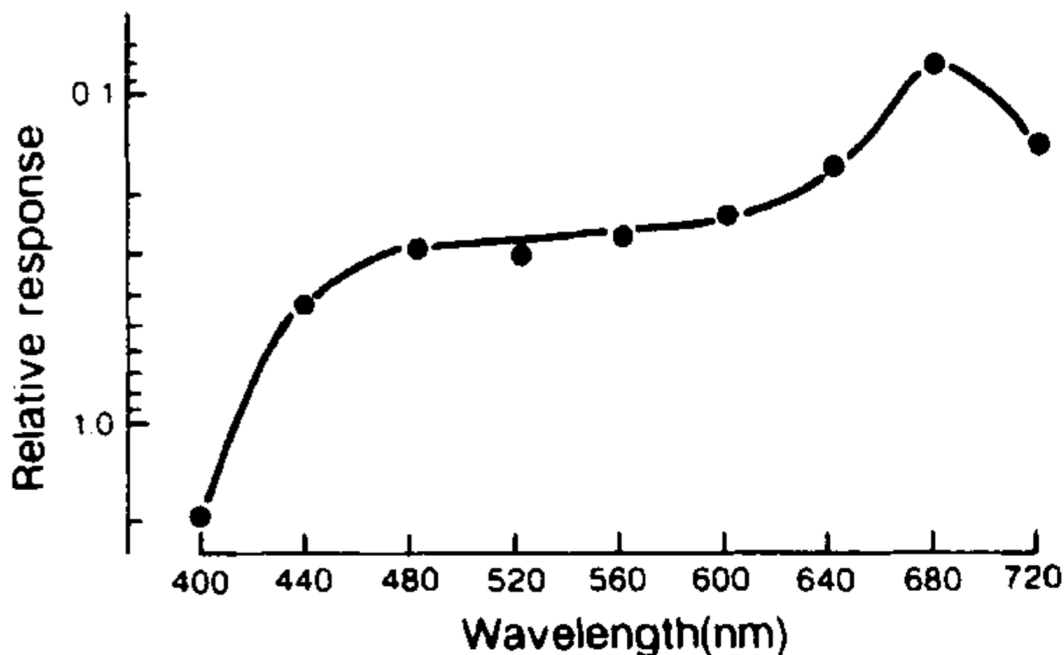


Figure 3. Action spectrum for photokinesis in *S. coeruleus*. The fluence rate of monochromatic light at nine different wavelengths which gave the half-maximal kinetic response (0.58 mm/s) was determined to obtain the action spectrum.

- **Y-axis is logarithmic *and* inverted.** The label reads “Relative response,” but the plotted units are actually fluence rate (W/m^2) needed to hit the 0.58 mm/s criterion velocity — same scale as Fig. 2a’s x-axis (log, ~0.1-1.0+ W/m^2). Neither Fig. 3’s axis nor its caption print a unit; W/m^2 is inferred by matching it against Fig. 2’s explicitly labeled intensity axis (“0.1 W/m^2 (400 nm)”, etc.), which measures the same quantity over the same numeric range. The axis is drawn inverted (small numbers near the top) so that “higher on the page” reads as “more sensitive” (less light needed) —

a standard action-spectrum convention, but it means the raw printed numbers are fluence rates, not sensitivity units.

- **Specific values:** 680 nm needed $\sim 0.08 \text{ W/m}^2$ to reach 0.58 mm/s (most sensitive point tested); 400 nm needed $\sim 1.1 \text{ W/m}^2$ (least sensitive) — a genuine $\sim 14\times$ ratio, not a mild bias. Curve shape: steep rise 400-440 nm, plateau $\sim 440\text{-}600$ nm, second climb to the peak at 680 nm, then a drop at 720 nm.
- **Why 0.58 mm/s as the criterion velocity:** the paper doesn't explain this number directly. It's *not* a measured dark-floor/light-ceiling midpoint — a true zero-light data point isn't measurable with their setup (the same light source doubles as the dark-field illumination the tracking camera needs to see the cell at all), and no plateau/saturation is described in either of their fluence-response figures across the *entire* tested range. The more likely explanation: 0.58 mm/s is a fixed criterion that happens to fall within the overlapping, actually-measured range of all nine wavelength-specific curves, letting a fluence-rate value be interpolated (not extrapolated) for each of the 9 wavelengths — the standard practical reason for picking one criterion response when collapsing several dose-response curves into a single action spectrum.